



CSEE4119: Computer Networks

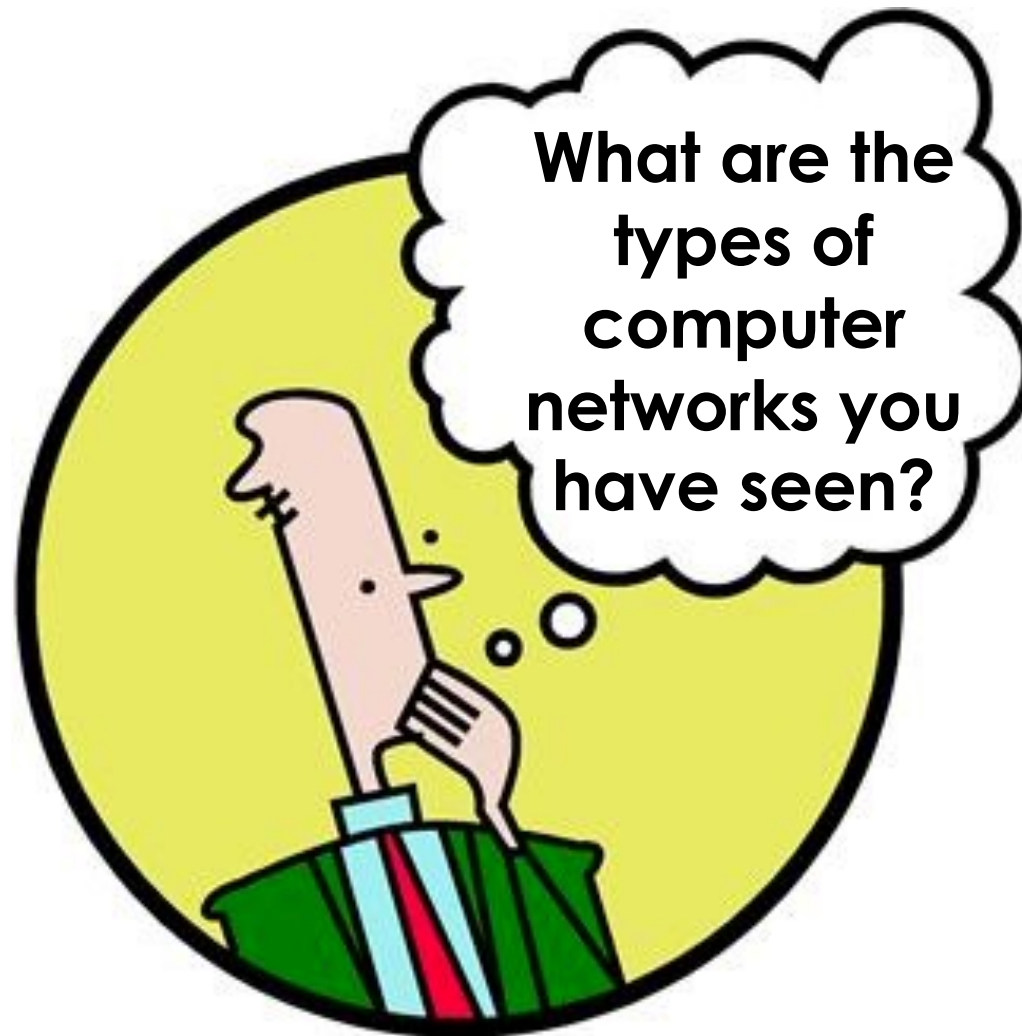
Xia Zhou
Columbia University
Jan. 20, 2026

Why Study Computer Networking?

#1: Networking is ubiquitous

- We use the Internet everyday, in many ways

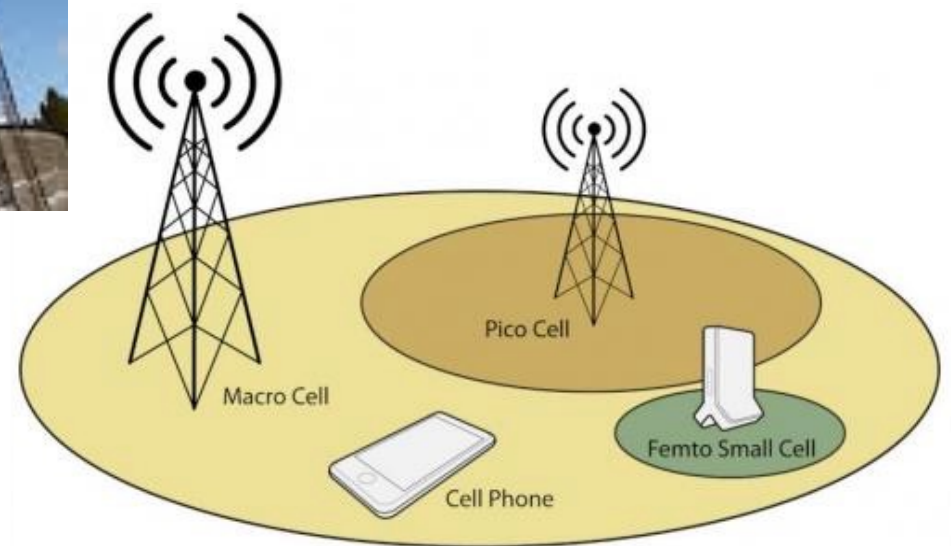




Wi-Fi Networks



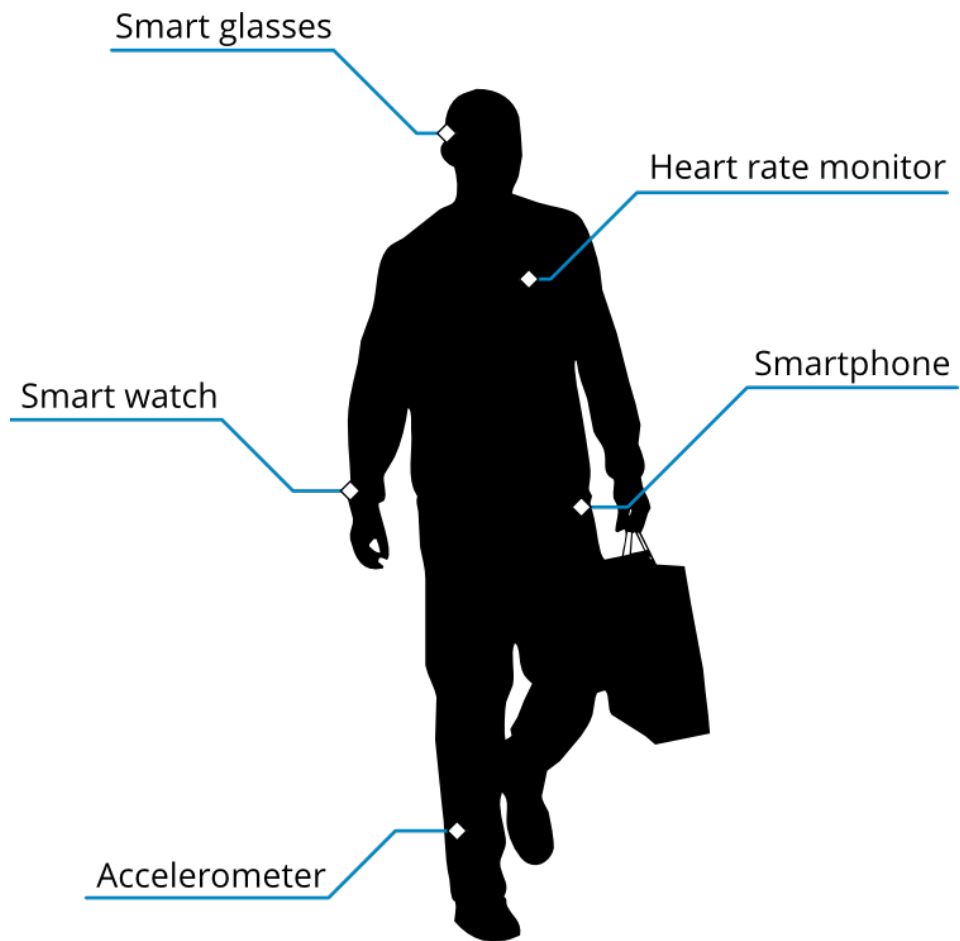
Cellular Networks



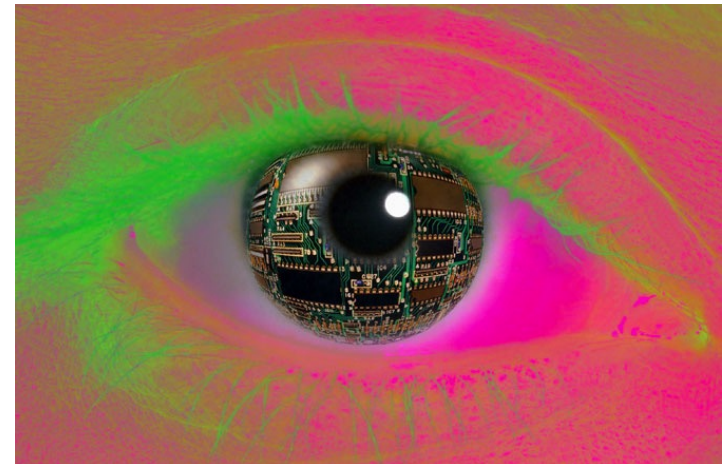
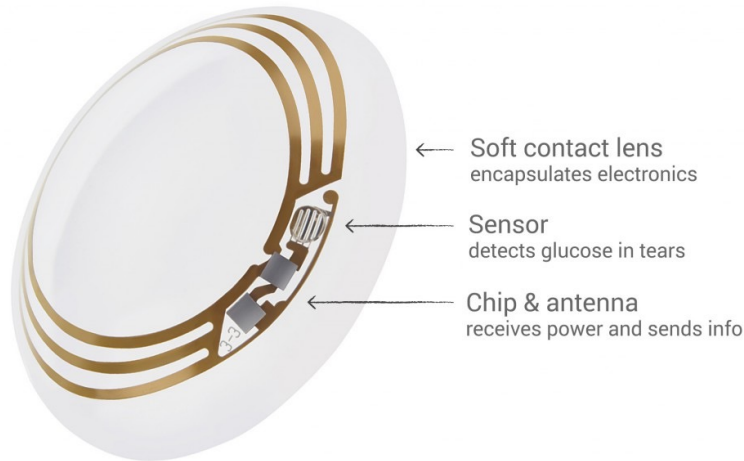
Data Center Networks



Body Area Networks

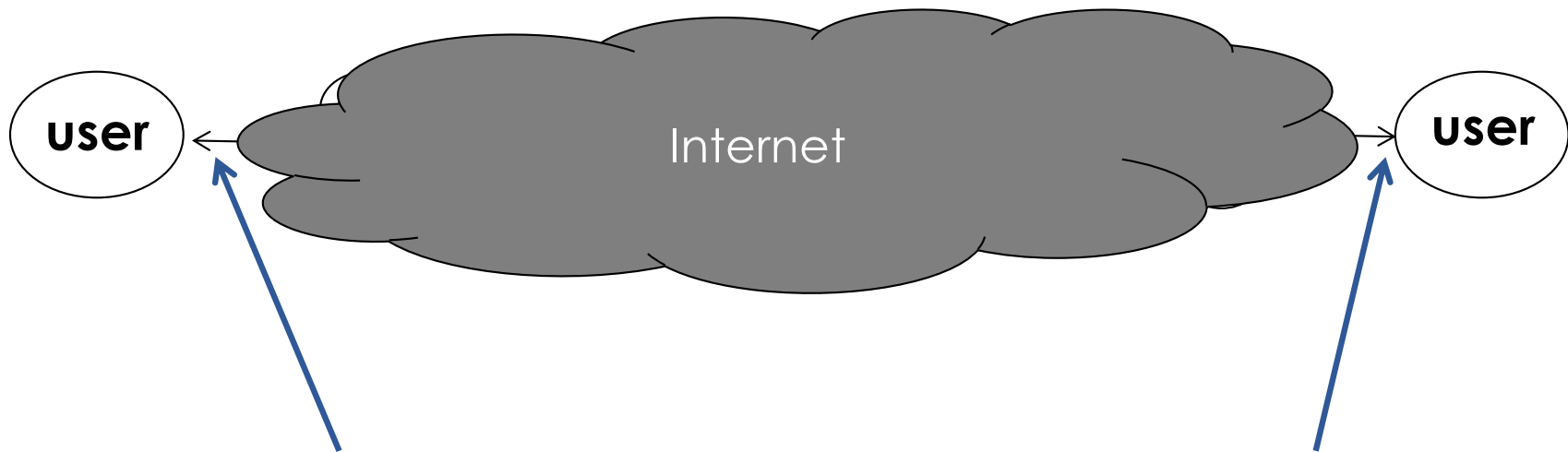


Computers Blended into Everyday Life



The Internet

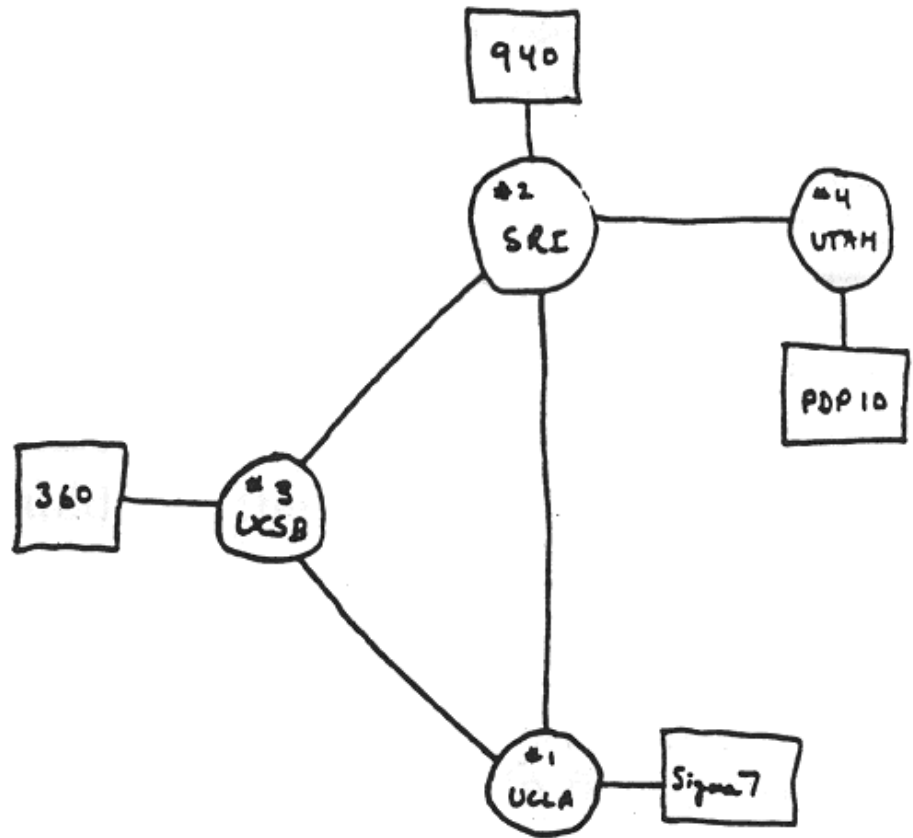
- Ties together different networks



Tied together by the “Internet Protocol” (IP): a single common interface between users and the network, and between networks

Fun Fact about the Internet

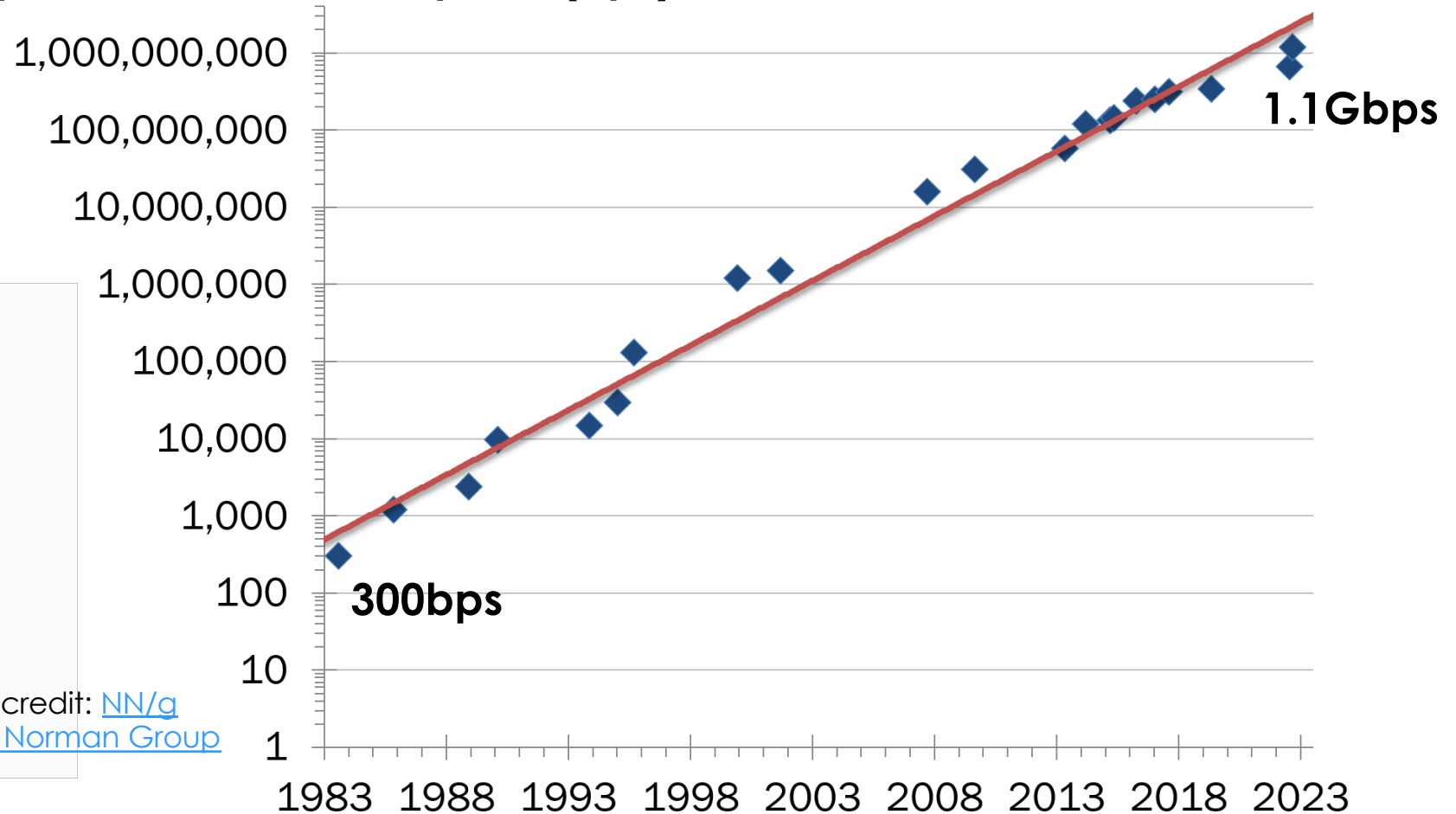
- Started out as a military project: ARPANET
- First node of ARPANET is operational in 1969
 - 15 nodes
 - First e-mail program



THE ARPA NETWORK

Tremendous Growth

Top-end user Internet speed (bps)



Picture credit: [NN/g](#)
[Nielsen Norman Group](#)

Nielsen's law of Internet bandwidth: A high-end user's connection speed grows by **50%** per year

Tremendous Scale

- **6 billion** users (74% of world population)
- **1.3 billion** websites
- **376 billion** emails sent and received per day
- **5.8 billion** mobile users
- **3 billion** Facebook monthly active users
- **1 billion hours** of YouTube video watched per day
- Video occupies **82%** of the Internet traffic in 2025

Why Study Computer Networking?

#2: The Internet is a critical infrastructure

- Highlighted by the pandemic, used by people to work, to socialize, to learn, to entertain, and more...
- Even in the age of AI, a critical infrastructure for making powerful AI possible
 - OpenAI relies on Microsoft Azure's cloud infrastructure

What We Will Learn

- **Knowledge: how the Internet works**

- Internet architecture
- Networking protocols
- Applications (Web, DNS, ...)

- **Skill: network programming**

- Socket programming
- Network protocol/application implementation and design

- **Insight: key principles/concepts in networking**

- Layers
- Protocols
- Naming

Topics

■ Network applications

- How common Internet applications work (e.g., web, peer-to-peer applications, DNS, video streaming)
- How to build applications that span multiple computers and use Internet to communicate

■ Network communication protocols

- Transport protocols (how to deliver message across Internet), network layer (how to send data from source to destination), and link layer (medium access control), physical layers

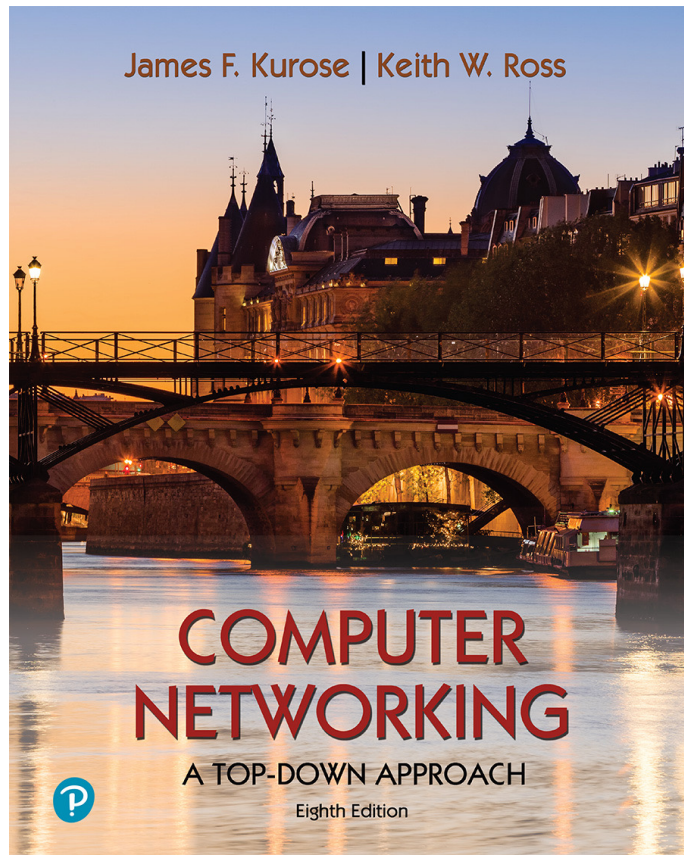
■ Advanced topics

- Data center networking, software-defined networking

How We Will Study Computer Networking

- Classroom time
 - Lectures
 - Join office hours if any questions
- Course work
 - Assignments
 - Written exams
 - Class project
 - In-class quizzes (informal)

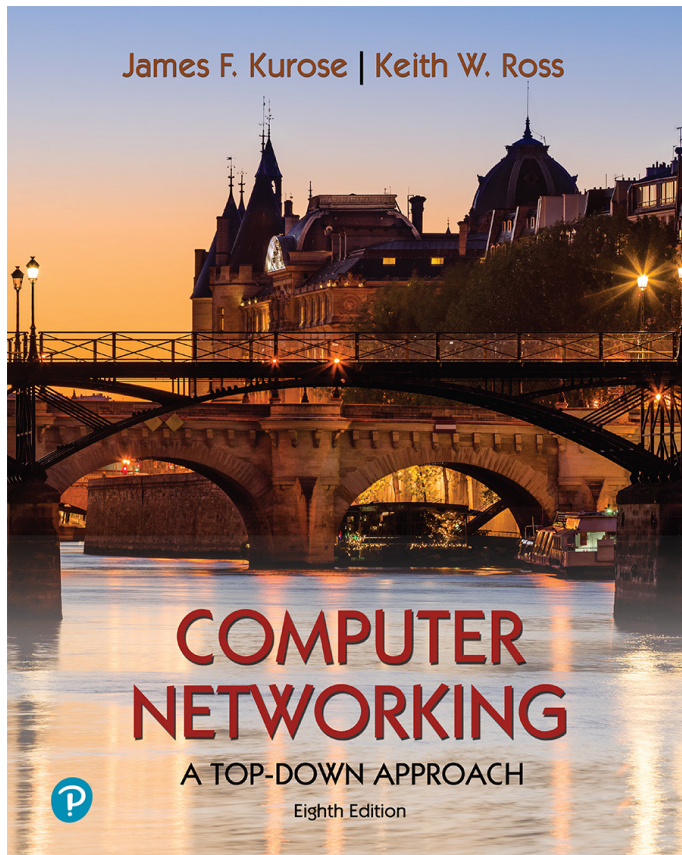
Textbook



*Computer Networking:
A Top-Down Approach*
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

- Authors went to Columbia!
- Focus on principle and practice that supported the tremendous Internet growth
- Well-organized supplementary materials (slides online)
- 7th/8th edition includes software defined networking

Topics We Will Cover



Computer Networking: A Top-Down Approach

8th edition

Jim Kurose, Keith Ross
Pearson, 2020

Chapters 1 - 7

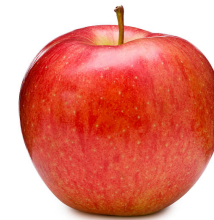
- Computer Networks Introduction
- Application Layer
- Transport Layer
- Network Layer: Data Plane
- Network Layer: Control Plane
- Link Layer
- Wireless and Mobile

We will not cover

- Security in Computer Networks
- Multimedia Networking

Grading

- 5%: Class participation and quizzes
 - Class discussions, Q/A
 - Occasional quizzes
 - Apple Quiz!
 - I will pose a question related to course material in class, whoever gives the best/correct answer wins an APPLE!



Grading

- 5%: Class participation and quizzes
- 40%: Assignments
 - Three assignments covering application, transport, and network layer protocols
- Submit on Gradescope
- Grades, grading feedback/comments will be released on CourseWorks

Submit Assignments On time

- No extension
- Because you do not need it
- Late submission (35% reduction per day)

Honor Code

- You can discuss high-level concepts and design ideas
- *Cannot code up labs together*
- *Cannot help others on coding*
- *Cannot share code snippets*
- *Cannot post your submission at any public repos*
- We have tools to detect code similarity
- *Check the honor code on CourseWorks*
- If anything is unclear at anytime, **ask** first

Policy of the Use of AI

- Permitted for deeper learning of concepts and protocols
- ***NOT permitted to write coding assignments or project***
 - Defeating the purpose of learning
- The work you submit should reflect your own work



**DO NOT SHARE
SLIDES AND CLASS MATERIALS
ON ONLINE SITES**
Course Hero

Uploading course materials to sites such as CourseHero, Chegg, or Github is academic misconduct at Columbia.

Grading

- 5%: Class participation and quizzes
- 40%: Assignments
- 20%: Team project
 - Last two weeks: no lectures, only invited talks
 - 3-4 students per team, each team shepherded by a TA
 - Design and code review
 - Final project demo with your shepherd

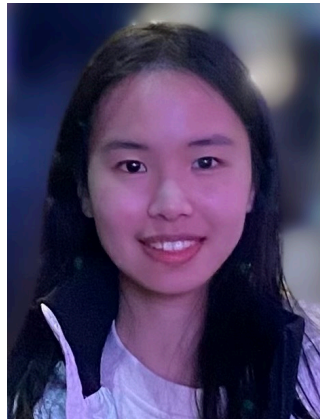
Grading

- 5%: Class participation and quizzes
- 40%: Assignments
- 20%: Team project
- 15%: Midterm
 - One class period on March 10
- 20%: Final exam (in-person)
 - May 14, Thursday, 4:10-7pm

Meet the TA Team!



Mike Azure



Siying Ding



Ananya Kapoor



Dev Bahl



Harris Ransom



Xiaofeng Yan



Ziang (Lex) Ren

Ask for Help

- Post on Ed Discussion
- Come to office hours
 - Xia's office hour in CEPSR 626 (1:30pm-2:30pm Thursday)
 - TA's office hours in EE lounge (check Courseworks for the schedule)
- Email:
 - To reach all TAs:
csee4119s26-ta@lists.cs.columbia.edu
 - To reach me alone: xia@cs.columbia.edu (put [CSEE4119] in the subject line)

Get Started

- Join CourseWorks
- Join Ed Discussion